# Arithmetic expression outside of simple scalar multiplication is currently n

https://github.com/tensorflow/quantum/issues/565

ROW 33

# Arithmetic expression outside of simple scalar multiplication is currently not supported #565

New issue

Closed

**theRoughCode** opened on May 7, 2021 · edited by theRoughCode

Edits

I'm trying to simulate the bias and weights in a classical neuron using `sympy` as follows:

```
qubits = cirq.GridQubit.rect(1, 1)
out_bits = qubits[-1:]
readouts = [cirq.Z(bit) for bit in out_bits]

encoding_params = sympy.symbols('enc0:{}'.format(self.input_dims))
weights = sympy.symbols('weights0:{}'.format(self.num_weights))
bias = sympy.symbols('bias0:{}'.format(self.num_bias))
params = [b + w * x for x, w, b in zip(encoding_params, weights[:3], bias[:3])]

circuit =cirq.Circuit(
      cirq.rx(params [0])(qubits[0]),
      cirq.ry(params [1])(qubits[0]),
      cirq.rz(params [2])(qubits[0])
  )


pqc = tfq.layers.ControlledPQC(circuit, readouts)
```

However, I'm getting the following error: `ValueError: Arithmetic expression outside of simple scalar multiplication is currently not supported. See serializer.py for more information.`

It seems like this sort of arithmetic manipulation of symbols is not supported yet. Is there a way around this?

**lockwo** on May 8, 2021 · edited by lockwo

Edits Contributor

I believe that comes from the fact that your params looks like `[bias0 + enc0*weights0, bias1 + enc1*weights1, bias2 + enc2*weights2]` and only multiplication by a constant is allowed e.g. `[10*weights0, 2*weights1, 4*weights2]`. You could probably get around this with a custom layer. This seems to be similar to what you are trying to achieve:

```
class bias_layer(tf.keras.layers.Layer):
    def __init__(self, num_params, readout_ops):
        super(bias_layer, self).__init__()
        self.w = tf.Variable(initial_value=np.random.uniform(0, 2 * np.pi, (1, num_params)), dtype="float32",
        self.b = tf.Variable(initial_value=np.random.uniform(0, 2 * np.pi, (1, num_params)), dtype="float32",
        self.layer = tfq.layers.ControlledPQC(circuit, readout_ops)

    def call(self, inputs):
        return self.layer([inputs, self.w + self.b])
```

**MichaelBroughton** on May 10, 2021

Collaborator

Thanks for jumping on this **@lockwo** ! Building on that point, we only support scalar multiplication in circuit symbols and not much else. If you want to Compute any kind of complicated expression involving your symbol before putting it into a circuit, you should do so in TF. i.e.:

```
my_variable = <some tf.Tensor>
step1 = tf.math.log(my_variable)
step2 = step1 ** 2 + 10
step3 = step2 / 3.14
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

Participants

Zotero

**MichaelBroughton** on May 10, 2021 · Collaborator

Thanks for jumping on this @lockwo ! Building on that point, we only support scalar multiplication in circuit symbols and not much else. If you want to Compute any kind of complicated expression involving your symbol before putting it into a circuit, you should do so in TF. i.e.:

```
my_variable = <some tf.Tensor>
step1 = tf.math.log(my_variable)
step2 = step1 ** 2 + 10
step3 = step2 / 3.14
other_values = <some other tf.Tensor>
step4 = tf.math.cos(other_values)

final_inputs = tf.concat([step3, step4], ...)
a_pqc = tfq.layers.ControlledPQC(...)([..., final_inputs])
```

The above will feed in values to the symbols in your circuit that have had those transformations applied beforehand. We didn't want to find ourselves re-implementing all this baisc math stuff on our end when TF can already do it very reliably and quickly for us.

Does this help clear things up ?

**MichaelBroughton** on Jun 8, 2021 · Collaborator

Going to close for now. Feel free to re-open if there's any more issues.

**MichaelBroughton** closed this as completed on Jun 8, 2021

**theRoughCode** on Jul 3, 2021 · Author

That definitely helps, sorry for the late response!

**lockwo** mentioned this on Jun 21, 2022

Arithmetic expression outside of simple scalar multiplication is currently not supported. #693

# Arithmetic expression outside of simple scalar multiplication is currently not supported. #693

Closed

New issue

**Gopal-Dahale** opened on Jun 21, 2022

I created a parameterized circuit using Cirq for Amplitude Encoding as described in Supervised Learning with Quantum Computers. The parameter is an arithmetic expression involving multiple operations like division and squaring. Below is the formula used. This value is used inside the $Ry$ gate.

$$\beta_j^g = 2\arcsin\left(\frac{\sqrt{\sum_{l=1}^{2g-1} |A_{(2j-1)2^{g-1}+l}|^2}}{\sqrt{\sum_{l=1}^{2g} |A_{(j-1)2^g+l}|^2}}\right).$$

Using the circuit in the `tfq.layers.ControlledPQC` gives the following error.

```
ValueError: Arithmetic expression outside of simple scalar multiplication is currently not supported.
See serializer.py for more information.
```

Any hack for this?

**lockwo** on Jun 21, 2022 — Contributor

Hard to say the best solution without seeing the exact code causing you trouble. But based on the error I would guess you are doing some disallowed symbol operations and that it could probably be done in a custom layer. I recommend #565 which had the same problem/error

**lockwo** on Aug 15, 2022 — Contributor

Any updates on this or should it be closed?

👍 1

**mhucka** on Dec 1, 2024 — Member

The OP appears to have given the thumbs-up to closing this issue, so I'm going ahead and closing it.

**mhucka** closed this as completed on Dec 1, 2024

**Assignees**
No one assigned

**Labels**
No labels

**Type**
No type

**Projects**
No projects

**Milestone**
No milestone

**Relationships**
None yet

**Development**
No branches or pull requests

**Notifications** Customize

Subscribe

You're not receiving notifications from this thread.

**Participants**